

PROBLEMA 2

1) $S \rightarrow OAO \mid 1B1 \mid BB$
 $A \rightarrow C$
 $B \rightarrow SIA$
 $C \rightarrow SIE$

④ Eliminar producciones ϵ

→ C tiene E

→ A y B pueden ser E

Reemply zur

$S \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB \mid B$
 $A \rightarrow \epsilon$ $B \rightarrow \epsilon$

P: S $\rightarrow$ 0A0 | 00 | 1B1 | 11 | BB | B
A $\rightarrow$ C
B $\rightarrow$ S | A
C $\rightarrow$ S

(b) Eliminar producciones unarias

(s, s) (A, A) (B, B) (C, C)

- (S, S) $S \rightarrow B$ (S, B)
- (S, B) $B \rightarrow A$ (S, A)
- (S, A) $A \rightarrow C$ (S, A)

• (S, C) $C \rightarrow S$ (S, S)

• (A, A) $A \rightarrow C$ (A, C)

• (A, C) $C \rightarrow S$ (A, S)

• (A, S) $S \rightarrow B$ (A, B)

• (B, B) $B \rightarrow A$ (B, A)

• (B, A) $A \rightarrow C$ (B, C)

• (B, C) $C \rightarrow S$ (B, S)

• (C, C) $C \rightarrow S$ (C, S)

• (C, S) $S \rightarrow B$ (C, B)

• (C, B) $B \rightarrow A$ (C, A)

Se puede llegar a cualquiera usando

$S \rightarrow B$ $B \rightarrow S \mid A$
 $A \rightarrow C$ $C \rightarrow S$

► Producciones no unarias

► (S, S) $S \rightarrow 0AC \mid 00 \mid 1B1 \mid 11 \mid BB$

► (C, S) $S \rightarrow 0AC \mid 00 \mid 1B1 \mid 11 \mid BB$
 $C \rightarrow 0AC \mid 00 \mid 1B1 \mid 11 \mid BB$

► (A, C) $C \rightarrow 0AC \mid 00 \mid 1B1 \mid 11 \mid BB$
 $A \rightarrow 0AC \mid 00 \mid 1B1 \mid 11 \mid BB$

► (B, S) $S \rightarrow 0AC \mid 00 \mid 1B1 \mid 11 \mid BB$

► (B, A) $A \rightarrow$ lo mismo
 $B \rightarrow$ lo mismo

P:

$$\begin{aligned}
 S &\rightarrow 0AO \mid 00 \mid 1B1 \mid 11 \mid BB \\
 A &\rightarrow 0AO \mid 00 \mid 1B1 \mid 11 \mid BB \\
 B &\rightarrow 0AO \mid 00 \mid 1B1 \mid 11 \mid BB \\
 C &\rightarrow 0AO \mid 00 \mid 1B1 \mid 11 \mid BB
 \end{aligned}$$

③ Eliminar símbolos inútiles

► $T = \{0, 1\}$ → símbolos terminales
se producen

$W_1 = \{S, A, B, C\}$ no hay inútiles

► $Y_1 = \{S\}$
se derivan

$Y_2 = \{A, B, S\}$

no se llega a C, es inalcanzable y se elimina

P:

$$\begin{aligned}
 S &\rightarrow 0AO \mid 00 \mid 1B1 \mid 11 \mid BB \\
 A &\rightarrow 0AO \mid 00 \mid 1B1 \mid 11 \mid BB \\
 B &\rightarrow 0AO \mid 00 \mid 1B1 \mid 11 \mid BB
 \end{aligned}$$

④ Forma normal de Chomsky

► Crear variables para símbolos terminales

$Y \rightarrow 0$

$Z \rightarrow 1$

► Reemplazar

$S \rightarrow YAY \mid YY \mid ZBZ \mid ZZ \mid BB$

$A \rightarrow YAY \mid YY \mid ZBZ \mid ZZ \mid BB$

$B \rightarrow YAY \mid YY \mid ZBZ \mid ZZ \mid BB$

ux

$$C_1 = AY$$

$$C_2 = BZ$$

P:

$$S \rightarrow YC_1 \mid YY \mid ZC_2 \mid ZZ \mid BB$$

$$A \rightarrow YC_1 \mid YY \mid ZC_2 \mid ZZ \mid BB$$

$$B \rightarrow YC_1 \mid YY \mid ZC_2 \mid ZZ \mid BB$$

$$Y \rightarrow 0$$

$$Z \rightarrow 1$$

$$C_1 \rightarrow AY$$

$$C_2 \rightarrow BZ$$

2)

$$S \rightarrow aAa \mid bBb \mid \epsilon$$

$$A \rightarrow C \mid a$$

$$B \rightarrow C \mid b$$

$$C \rightarrow CDE \mid \epsilon$$

$$D \rightarrow A \mid B \mid ab$$

@ Eliminar ϵ

- S tiene ϵ pero no tiene "producción propia"
- C tiene ϵ
- A y B pueden ser ϵ porque producen C
- D puede ser ϵ porque produce A y B

► Reemplazar

$$\rightarrow S \rightarrow aAa \mid a \mid bBb \mid b$$

$$A \rightarrow \epsilon$$

$$B \rightarrow \epsilon$$

$$A \rightarrow C \mid a$$

$$B \rightarrow C \mid b$$

$$C \rightarrow CDE \mid DE \mid CE \mid \epsilon$$

$$C \rightarrow \epsilon$$

$$D \rightarrow \epsilon$$

ambos

$$D \rightarrow A \mid B \mid ab$$

⑥ Eliminar producciones unarias
 (S, S) (A, A) (B, B) (C, C) (D, D)
 y notarse

• (A, A) $A \rightarrow C$ (A, C)

• (A, C) $C \rightarrow E$ (A, E)

• (B, B) $B \rightarrow C$ (B, C)

• (B, C) $C \rightarrow E$ (B, E)

• (C, C) $C \rightarrow E$ (C, E)

• (D, D) $D \rightarrow A$ (D, A)

• (D, A) $A \rightarrow C$ (D, C)

• (D, C) $C \rightarrow E$ (D, E)

• (D, D) $D \rightarrow B$ (D, B)

→ Producciones no unarias

• (C, C) $C \rightarrow CDE \mid DE \mid CE$

• (A, C) $A \rightarrow CDE \mid DE \mid CE \mid a$

• (B, C) $B \rightarrow CDE \mid DE \mid CE \mid b$

• (D, A) $D \rightarrow CDE \mid DE \mid CE \mid a \mid B \mid ab$

• (D, B) $D \rightarrow CDE \mid DE \mid CE \mid a \mid b \mid ab$

P: $S \rightarrow a A a \mid a a \mid b B b \mid b b$

$A \rightarrow CDE \mid DE \mid CE \mid a$

$B \rightarrow CDE \mid DE \mid CE \mid b$

$C \rightarrow CDE \mid DE \mid CE$

$D \rightarrow CDE \mid DE \mid CE \mid a \mid b \mid ab$

© Eliminar símbolos inútiles e inalcanzables

► $T = \{a, b\}$

$W_1 = \{S, A, B, D\}$

$W_2 = \{S, A, B, D, C\}$ produce

E es un símbolo inútil

Como todas las alternativas de C requieren E lo quitamos

P: $S \rightarrow aAa \mid aa \mid bBb \mid bb$
 $A \rightarrow a$
 $B \rightarrow b$
 $D \rightarrow a \mid b \mid ab$

► $Y_1 = \{S\}$

lo deriva

$Y_2 = \{A, B, S\}$: no se deriva nada mas

D no aparece, es inalcanzable y se quita

P: $S \rightarrow aAa \mid aa \mid bBb \mid bb$
 $A \rightarrow a$
 $B \rightarrow b$

© Forma normal de Chomsky

► Crear variables para símbolos terminales

$Y \rightarrow a$

$Z \rightarrow b$

Reemplazar

$S \rightarrow YAY \mid YY \mid ZBZ \mid ZZ$

$A \rightarrow Y$

$B \rightarrow Z$

$C_1 \equiv AY$

$C_2 \equiv BZ$

P: $S \rightarrow YC_1 \mid YY \mid ZC_2 \mid ZZ$

$A \rightarrow Y$

$B \rightarrow Z$

$C_1 \rightarrow AY$

$C_2 \rightarrow BZ$

3)

EXERCICIO

$$S \rightarrow ASA | aB$$

$$A \rightarrow B | S$$

$$B \rightarrow b | \epsilon$$

1. Eliminar epsilon

$$S \rightarrow ASA | a | aB$$

$$A \rightarrow B | S$$

$$B \rightarrow b$$

$$\rightarrow A \rightarrow \epsilon | S$$

$$P: S \rightarrow ASA | a | AS | SA | S | aB$$

$$A \rightarrow B | S$$

$$B \rightarrow b$$

2. Unarias

(S, S) (A, A) $n)$

$P(A, A)$ $(A \rightarrow S)$ (A, S)

(A, B) $A \rightarrow B$ (A, B)

Producciones unarias

(A, S) $S \rightarrow ASA | a | AS | SA | S$

$A \rightarrow S | ASA | a | AS | SA | S | aB$

(A, B) $B \rightarrow b$

$A \rightarrow S | ASA | a | AS | SA | S | aB | b$

P:

$S \rightarrow A | A | AS | SA | S | aB | a$

$A \rightarrow S | A | A | a | AS | SA | S | aB | b$

$B \rightarrow b$

3. inútiles e indeseables

$$T = \{a, b\}$$

$$W_1 = \{S, A, B\}$$

$$W_2 = \{S, A, B\}$$

no hay
inútiles

$$Y_1 = \{S\}$$

$$Y_2 = \{S, A, B\}$$

$$Y_3 = \{S, A, B, a, b\}$$

no hay
indeseables

4. Chomsky

$$Y \rightarrow a$$

$$Z \rightarrow b$$

1. Remplazar

$$S \rightarrow ASA | SA | AS | S | YZ | Y$$

$$A \rightarrow ASA | SA | AS | S | YZ | Y | Z$$

$$B \rightarrow Z$$

$$C_1 = SA$$

$$R // S \rightarrow AC_1 | C_1 | AS | S | YZ | Y$$

$$A \rightarrow AC_1 | C_1 | AS | S | YZ | Z$$

$$B \rightarrow Z$$

$$Y \rightarrow a$$

$$Z \rightarrow b$$

$$C_1 \rightarrow SA$$